

AnkiDroid BANKAI — Technical Specification & SRS Guide

Production Spaced Repetition Engine • SuperMemo-2 • FSRS • Analytics Dashboard

EXECUTIVE SUMMARY

AnkiDroid BANKAI is a high-performance, open-source Android flashcard learning system built with Jetpack Compose and modern Kotlin. It provides dual-algorithm spaced repetition (Classic SM-2 and modern Free Spaced Repetition Scheduler - FSRS), a 60-day visual Review Heatmap, Anki-compliant Cloze Deletions, in-study microphone voice recording for pronunciation comparison, colored card flags, swipe gestures, and built-in SQLite database integrity diagnostics and automated snapshot backups.

1. SRS SCHEDULING ENGINES & ALGORITHMS

- SuperMemo 2 (SM-2):** The battle-tested Anki default interval scheduling algorithm with starting ease 2.5, dynamic ease adjustments (+0.15 for Easy, -0.15 for Hard, -0.20 for Again), and graduated intervals.
- FSRS (Free Spaced Repetition):** Modern mathematical retention model based on DSR (Difficulty, Stability, and Retrievability). Accurately schedules cards to hit an exact targeted retention rate (e.g. 90%).
- FSRS Weight Optimization:** Computes personalized decay weights dynamically from the user's historical review logs.
- Dynamic Fuzz Factor:** Applies pseudo-random interval variance ($\pm 5\%$) to prevent review clumping.
- Custom Learning & Relearning Steps:** Configurable intra-day and inter-day step progression (e.g. 1m, 10m, 1d).
- Lapse & Leech Thresholds:** Auto-identifies persistent failure cards (≥ 8 lapses) as 'Leeches' with warning alerts.

2. COMPREHENSIVE STUDY ANALYTICS & DASHBOARD

- 60-Day Review Heatmap:** GitHub-style visual activity matrix with 5 intensity levels tracking study consistency.
- Card Maturity Breakdown:** Categorizes collection into Mature ($\geq 21d$), Young ($< 21d$), Learning, New, and Suspended.
- Scheduling Forecast:** Projects upcoming card loads for Today, Tomorrow, 2-3 Days, 4-7 Days, and 8-30 Days.
- Retention & Accuracy KPIs:** Measures historical pass/fail recall percentages across all study logs.
- Active Study Time:** Computes total minutes engaged based on logged reviews.

3. ADVANCED CARD FORMATTING & AUDIO INTEGRATION

- Cloze Deletion Parsing:** Full support for standard Anki `{{c1::answer}}` and `{{c1::answer::hint}}` syntax.
- Type-in-the-Answer Mode:** Interactive text verification comparing spelling, typos, and German umlauts (ä, ö, ü, ß).
- Dual-Direction Study:** Seamless switching between German → English, English → German, or Alternating mixed sets.
- Pronunciation Mic Tester:** Built-in microphone recorder (AAC/MPEG4) with instant playback vs native German TTS.
- Dynamic Synonyms & AI:** Gemini AI German vocabulary assistant providing contextual synonyms and example sentences.

4. NAVIGATION, GESTURES & HAPTIC CONTROLS

- **Swipe Gestures:** Swipe Right to mark 'Good', Swipe Left to mark 'Again', Swipe to reveal answer.
- **Haptic Feedback:** Tactile vibration confirmation on grading button presses via LocalHapticFeedback.
- **Built-in Whiteboard:** Full-screen finger/stylus canvas overlay for practicing written German words and umlauts.
- **Undo Facility:** One-tap restoration of the previous card's exact SRS state and interval.
- **Card Diagnostics:** Deep inspection dialog displaying interval, ease factor, lapses, due date, and history log.
- **Four-Color Flagging:** Mark cards during study with Red, Orange, Green, or Blue flags for targeted review.
- **Timeboxing Timer:** Configurable study interval alarm prompting periodic breaks to prevent burnout.

5. CARD BROWSER & ANKI SEARCH SYNTAX

- **Anki Query Syntax:** Search by 'is:leech', 'is:suspended', 'is:new', 'is:learning', 'is:review', 'is:flagged'.
- **Color Flag Filtering:** Query by exact flag: 'flag:red', 'flag:orange', 'flag:green', 'flag:blue'.
- **Multi-Select Batch Actions:** Batch delete, batch flag, batch suspend/unsuspend, and batch reset SRS progress.
- **Micro-Set Range Filter:** Slice decks into 25-card micro-batches for focused study sessions.
- **Flexible Sorting:** Sort by order index, alphabetically, due date, lowest ease, most lapses, or flags first.

6. DATABASE INTEGRITY & AUTOMATED SNAPSHOT BACKUPS

- **PRAGMA integrity_check:** Direct SQLite integrity scan verifying B-tree pages, indices, and constraints.
- **Vacuum & Analyze Tool:** Compacts disk storage, reclaims unused space, and re-optimizes query plans.
- **Automated JSON Backups:** One-touch export generating complete standalone snapshots of decks, cards, and logs.
- **Migration Pipeline:** Clean Room Database v1 → v2 migration preserving all user data and study progress.
- **Anki .apkg & CSV Importer:** Full support for importing standard Anki packages and customizable CSV files.

7. SYSTEM ARCHITECTURE & COMPONENTS

LAYER	IMPLEMENTATION DETAILS
UI & Presentation	100% Jetpack Compose M3, Adaptive Screens, Custom Canvas Whiteboard & Heatmap
State Management	MVVM Architecture with StateFlow, collectAsStateWithLifecycle, and Coroutines
Local Persistence	Room Database 2.6 (SQLite), multi-table relationships, migration scripts
SRS Math Engine	AnkiSrsEngine (SM-2) + FsrEngine (DSR model) with customizable deck settings
Audio & Hardware	Android TextToSpeech, MediaRecorder (AAC/MPEG4), MediaPlayer, Vibrator Haptics